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1957

Cleaning, Disinfection, and Sterilization

EMS 313 / Chapter 4

Public Health and Safety

Learning objectives

- Define cleaning, disinfection, sterilization, and decontamination.
- Classify equipment by the body area it contacts.
- Explain the steps of effective cleaning.
- Compare disinfectant levels and common chemical agents.
- Describe sterilization methods and causes of failure.
- Apply these principles to equipment used in EMS.

Infection risk from EMS equipment

- Used equipment can carry pathogens to the next patient.
- Hands can move pathogens between equipment and people.
- Cleaning removes contamination. Disinfection kills remaining susceptible organisms.
- Include stretcher rails, monitor controls, cables, and door handles.



Stretcher rails and handles can transfer organisms to hands.

Key definitions

Term	Meaning	Example
Cleaning	Removes soil and many microorganisms by physical action.	Detergent and friction
Disinfection	Destroys many or all pathogens, usually excluding bacterial spores.	Approved surface disinfectant
Sterilization	Destroys all microbial life, including bacterial spores.	Validated steam cycle
Decontamination	Makes an item safe to handle, use, or discard.	A combination of appropriate methods

The required method depends on how the item will be used.

Antiseptics and disinfectants

Agent	Where it is used	Example
Antiseptic	Living tissue, especially skin	A skin preparation before IV insertion
Disinfectant	Equipment and environmental surfaces	A product for a stretcher or monitor

The chemical name alone does not tell you the correct use. Check the product label.

Microorganisms and disinfectant terms

Term	Simple meaning
Vegetative bacteria	Bacteria in their growing, active form
Bacterial spore	A resistant survival form produced by some bacteria
Bactericidal / virucidal / fungicidal	Kills bacteria / viruses / fungi
Tuberculocidal	Kills Mycobacterium tuberculosis under the stated conditions
Sporicidal	Kills bacterial spores under the stated conditions

A product that kills ordinary bacteria may still leave bacterial spores.

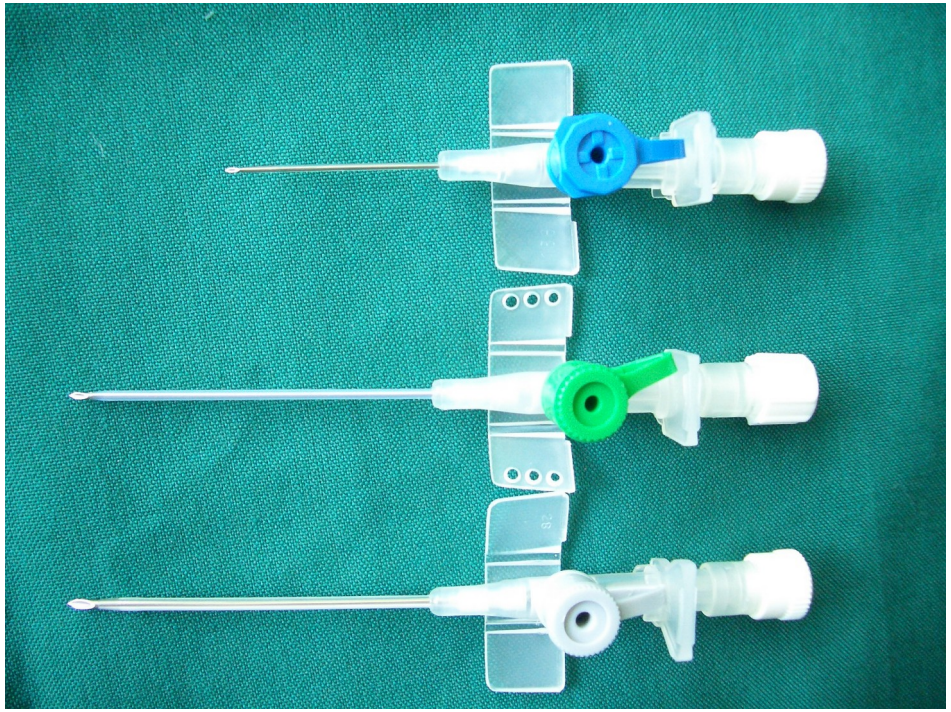
Spaulding classification

Item class	Body contact	Minimum processing	EMS example
Critical	Sterile tissue or the bloodstream	Must be sterile	Single-use IV cannula
Semicritical	Mucous membranes or non-intact skin	High-level disinfection	Reusable laryngoscope blade
Noncritical	Intact skin	Low-level disinfection	Blood-pressure cuff

Clean reusable items before the required disinfection or sterilization process.

Critical items

- Critical items enter sterile tissue or the bloodstream.
- Examples include intravenous (IV) cannulas and intraosseous (IO) needles.
- They must be sterile before use.
- Check the sealed package, expiry date, and single-use label.



IV cannulas enter the bloodstream and must be sterile.

Photo: Saltanat ebli / CC0

Discard single-use devices after use. Do not use a wet or damaged sterile package.

Semicritical items

- Semicritical items contact mucous membranes or broken skin.
- Examples include reusable laryngoscope blades and some endoscopes.
- Clean first, then apply at least high-level disinfection.
- A surface wipe alone does not meet this requirement.



The laryngoscope blade contacts the lining of the mouth and airway.

Photo: Welch Allyn, via McKesson

Follow the blade’s instructions. They may require sterilization.

Noncritical equipment and surfaces

- Noncritical items contact intact skin.
- Examples include blood-pressure cuffs and stethoscopes.
- Clean and disinfect reusable equipment between patients as directed.
- Use a compatible product and the full labelled contact time.



Cuffs and stethoscopes usually need low-level disinfection.

Photo: CDC, via Wikimedia Commons / Public domain

Blood or specific pathogens may require a different disinfectant.

Cleaning before disinfection

- Blood and tissue can shield microorganisms.
- Organic material can also reduce disinfectant activity.
- Water, detergent, and friction remove contamination.
- Clean promptly. Dried material is harder to remove.



Brushing helps remove soil from instrument surfaces and joints.

Photo: Santé Québec, Centre-Sud-de-l'Île-de-Montréal

Enzymatic cleaners help break down organic material. Disinfection is still required.

Steps in cleaning reusable equipment

1. Preparation

Wear appropriate PPE. Separate dirty equipment from clean supplies.

2. Disassembly

Open joints and separate removable parts as the instructions allow.

3. Cleaning

Use the specified detergent and friction. Reach hinges and accessible channels.

4. Rinsing and inspection

Remove cleaning residue and check for remaining soil or damage.

5. Next stage

Proceed to the required disinfection or sterilization, then dry and store as directed.

Levels of disinfection

Level	Expected action	Typical application
Low-level	Kills most vegetative bacteria, some fungi, and some viruses.	Noncritical equipment
Intermediate-level	Also kills mycobacteria and most viruses and fungi. Does not reliably kill spores.	Selected contaminated surfaces
High-level	Kills all microorganisms except potentially large numbers of bacterial spores.	Semicritical equipment

Sterilization destroys all microbial life, including spores, through a validated process.

Contact time for surface disinfection

- Read the product label before use.
- Apply enough disinfectant to wet the required surface.
- Keep it wet for the full labelled contact time.
- If it dries early, reapply as the label directs.



The surface must remain wet for the time stated on the label.

Photo: Health Care Logistics

Example only: a 3-minute contact time means 3 full minutes of wet contact.

Selection of a disinfectant

- Identify the item class and the required level of disinfection.
- Check the product’s claims against the expected pathogen.
- Confirm compatibility with the device material.
- Follow the labelled dilution, contact time, temperature, and use-life.
- Check required PPE, ventilation, rinsing, and storage.

A chemical name or a “kills 99.9%” claim does not establish suitability for every device.

Alcohols

- Examples are ethanol and isopropyl alcohol.
- They act rapidly against many bacteria, fungi, and viruses, but do not kill bacterial spores.
- Alcohol is unsuitable for sterilization or high-level disinfection of a laryngoscope blade.
- Rapid evaporation can prevent adequate contact time.
- Repeated use may damage some plastics and rubber. Alcohol is flammable.

Use only a formulation and application approved for the equipment.

Chlorine compounds

- Sodium hypochlorite is the active ingredient in many bleach products.
- Appropriate formulations act against a broad range of microorganisms.
- Services may use them for blood spills or specific pathogen contamination.
- Blood and other organic material can reduce activity.
- Chlorine can irritate skin and eyes, bleach fabric, and corrode metal.

Never mix bleach with acids, ammonia, or other cleaning chemicals.

Quaternary ammonium compounds

- These disinfectants are commonly called quats.
- They occur in many products for noncritical equipment and surfaces.
- Typical formulations act against vegetative bacteria and enveloped viruses.
- Basic quat formulations have limited activity against spores, mycobacteria, and many non-enveloped viruses.
- Combination products may have different claims. Check the label.

Enveloped viruses have an outer lipid layer. Non-enveloped viruses often resist more agents.

Agents for high-level disinfection

Agent	Main role	Important limitation
Glutaraldehyde	High-level disinfection with correct exposure. Active against mycobacteria.	Can fix protein to equipment. Irritating vapour.
Ortho-phthalaldehyde (OPA)	High-level disinfection of compatible reusable devices.	Can stain protein and irritate eyes. Limited sporicidal activity.
Peracetic acid systems	Controlled processing of compatible devices.	Strong oxidizer. Requires a validated system and cycle.

These agents require trained staff, appropriate PPE, and the complete processing instructions.

Hydrogen peroxide formulations

Formulation	Use	Key point
Hydrogen peroxide	Surface disinfection or high-level disinfection, depending on the product.	Concentration and labelled conditions determine the level.
Improved / accelerated hydrogen peroxide	Formulations with additives that improve antimicrobial action.	Claims and contact times vary by product.

An ordinary household peroxide bottle does not automatically qualify as a high-level disinfectant.

Other chemical agents

Agent	Main point	Limitation
Formaldehyde	An older disinfectant and chemical sterilant in specific applications.	Hazardous fumes and carcinogenicity limit its healthcare use.
Iodophors	Iodine combined with a carrier. Products exist for skin or surfaces.	Skin and surface formulations are not interchangeable.
Phenolics	Some products disinfect noncritical surfaces.	Unsuitable for high-level disinfection. Special precautions apply around neonates.

Use the formulation intended for the task, according to its label.

Other methods of microbial control

Method	How it works	Limitation
Antimicrobial metals	Some metal ions inhibit or kill microorganisms.	Specialized application. Does not replace routine processing.
Ultraviolet C (UV-C)	Controlled exposure can inactivate microorganisms.	Shadows and soil reduce action. Equipment and dose matter.
Pasteurization	Hot-water treatment reduces microorganisms.	Does not reliably destroy spores. It is not sterilization.

Additional technologies do not remove the need for the required cleaning process.

Endoscopes and internal channels

- A lumen is a channel inside a device.
- Blood, secretions, and microorganisms can remain in these channels.
- A bronchoscope contacts airway mucosa and needs at least high-level disinfection.
- Scopes that enter sterile tissue require sterilization.



A bronchoscope has a flexible insertion tube with internal channels.

Photo: Håkon Olav Leira / CC BY-SA 3.0

Cleaning must reach the channels. Disinfecting only the outside is insufficient.

Sterilization

- Sterilization destroys all forms of microbial life, including bacterial spores.
- Critical reusable devices require a validated sterilization process.
- Steam is preferred for compatible items that tolerate heat and moisture.
- Heat-sensitive devices need a compatible low-temperature method.
- Safe packaging and storage protect the item after processing.

A damaged or wet sterile package compromises safe use.

Steam sterilization: the autoclave

- An autoclave uses steam under pressure.
- Steam transfers heat and destroys microorganisms.
- Common cycles use 121°C or 132°C. The required time varies.
- Steam must reach every item during the validated cycle.



An open autoclave shows the chamber and trays used for instrument loads.

Photo: Midmark Corporation

Follow the cycle specified for the device, packaging, and load.

Low-temperature sterilization

Method	Typical application	Key consideration
Ethylene oxide (EtO)	Compatible items that cannot tolerate steam.	Toxic gas. Requires controlled processing and aeration.
Hydrogen peroxide vapour / gas plasma	Compatible heat-sensitive devices.	Material and lumen restrictions vary by system.
Liquid chemical sterilant systems	Selected immersible devices under validated conditions.	Rinsing and immediate-use requirements may limit storage.

Select the method from the device’s validated instructions.

Factors that reduce effectiveness

Factor	Why it matters
Soil or a high microbial load	Material shields organisms and may consume the chemical.
Hinges, crevices, and lumens	The agent may not reach every contaminated surface.
Resistant organisms or biofilm	Some organisms and protective layers resist the process.
Wrong concentration or expired solution	The product may no longer achieve its labelled effect.
Incorrect time, temperature, or pH	The process falls outside its validated working conditions.

Safe work practices

- Wear gloves and add gown, eye, and face protection when splash exposure is likely.
- Use ventilation required by the product and processing system.
- Keep chemicals labelled and follow their safety data sheets.
- Never mix chemicals or improvise concentrations.
- Separate dirty equipment from clean equipment, then perform hand hygiene.

Gloves for patient care may not provide adequate protection against every chemical.

EMS case: after a patient transport

Reusable laryngoscope blade

Secretions and dried blood remain around the hinge.

IV cannula package

The unopened package is wet after lying in a pool of fluid.

Blood-pressure cuff and stretcher

The patient vomited during transport.

For each item: identify the class, choose the action, and explain one possible failure.

EMS case: appropriate actions

Item	Appropriate action	Common failure
Reusable blade	Clean the hinge and all surfaces. Apply at least high-level disinfection, or sterilize as instructed.	Only wiping the visible surface
Wet cannula package	Remove it from use and discard the compromised single-use device.	Assuming a closed pack is still sterile
Cuff and stretcher	Remove vomit. Use a compatible approved product with the necessary pathogen claim and full contact time.	Wrong product or early drying

Follow device instructions, the product label, and the service's infection-control procedure.

Key points

- Classify equipment by the body area it contacts.
- Clean reusable items before the required disinfection or sterilization.
- Use a product with the appropriate level and pathogen claims.
- Follow the full labelled processing conditions.
- Protect clean and sterile equipment during storage and handling.

Critical: sterile. Semicritical: at least high-level disinfection. Noncritical: usually low-level disinfection.

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